

M T W T F S S
☐ ☐ ☐ ☐ ☐ ☐ ☐

Date: _____

Cryptology Theory - 4.

1. Decrypt the following ciphertext using the Hill cipher.

i) Cipher text: HLDTG IHTD XWQ CMAG, Key = $\begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix}$

use (A=0, B=1, ..., Z=25).

find the plain text.

Solⁿ

Inverse.

$$\begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix}^{-1} = (3 \times 7) - (5 \times 2) = 21 - 10 = 11$$

$$11^{-1} \pmod{26} \Rightarrow 26 = 11(2) + 4, \quad 4 = 26 - 11(2)$$

$$11 = 4(2) + 3, \quad 3 = 11 - 4(2)$$

$$4 = 3(1) + 1$$

$$1 = 4 - 3$$

$$1 = 4 - (11 - 4(2))$$

$$1 = 3(4) - 11$$

$$1 = 3(26 - 11(2)) - 11$$

$$1 = (-7)11 + 3(26), \quad -7 + 26 = 19$$

Decryption. $P = 11^{-1} C \pmod{26}$

$$19 \begin{bmatrix} 7 & -2 \\ -5 & 3 \end{bmatrix} \pmod{26} = \begin{bmatrix} 133 & -38 \\ -95 & 57 \end{bmatrix} \pmod{26} = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix}$$

$$\text{ii)} \quad \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 7 \\ 11 \end{bmatrix} \pmod{26} = \begin{bmatrix} 3 \times 7 + 14 \times 11 \\ 9 \times 7 + 5 \times 11 \end{bmatrix} \pmod{26} = \begin{bmatrix} 175 \\ 118 \end{bmatrix}$$

$$\pmod{26} = \begin{bmatrix} 19 \\ 14 \end{bmatrix}$$

$$\text{iii)} \quad \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 3 \\ 19 \end{bmatrix} \pmod{26} = \begin{bmatrix} 3 \times 3 + 14 \times 19 \\ 9 \times 3 + 5 \times 19 \end{bmatrix} \pmod{26}$$

$$\begin{bmatrix} 275 \\ 122 \end{bmatrix} \pmod{26} = \begin{bmatrix} 15 \\ 18 \end{bmatrix}$$

$$\text{iii)} \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 16 \\ 8 \end{bmatrix} \pmod{26} = \begin{bmatrix} 3 \times 16 + 14 \times 8 \\ 9 \times 16 + 5 \times 8 \end{bmatrix} \pmod{26}$$

$$\begin{bmatrix} 160 \\ 184 \end{bmatrix} \pmod{26} = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$$

$$\text{iv)} \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 7 \\ 9 \end{bmatrix} \pmod{26} = \begin{bmatrix} 3 \times 7 + 14 \times 9 \\ 9 \times 7 + 5 \times 9 \end{bmatrix} \pmod{26}$$

$$\begin{bmatrix} 147 \\ 108 \end{bmatrix} \pmod{26} = \begin{bmatrix} 17 \\ 4 \end{bmatrix}$$

$$\text{v)} \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 3 \\ 23 \end{bmatrix} = \begin{bmatrix} 3 \times 3 + 14 \times 23 \\ 9 \times 3 + 5 \times 23 \end{bmatrix} \pmod{26}$$

$$= \begin{bmatrix} 831 \\ 142 \end{bmatrix} \pmod{26} = \begin{bmatrix} 19 \\ 12 \end{bmatrix}$$

vi) =

$$\begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 22 \\ 16 \end{bmatrix} = \begin{bmatrix} 3 \times 22 + 14 \times 16 \\ 9 \times 22 + 5 \times 16 \end{bmatrix} \pmod{26} = \begin{bmatrix} 290 \\ 278 \end{bmatrix}$$

$$\pmod{26} = \begin{bmatrix} 4 \\ 18 \end{bmatrix}$$

$$\text{vii)} \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 2 \\ 12 \end{bmatrix} = \begin{bmatrix} 3 \times 2 + 14 \times 12 \\ 9 \times 2 + 5 \times 12 \end{bmatrix} \pmod{26} = \begin{bmatrix} 174 \\ 78 \end{bmatrix}$$

$$\pmod{26} = \begin{bmatrix} 18 \\ 0 \end{bmatrix}$$

$$\text{viii)} \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 0 \\ 6 \end{bmatrix} = \begin{bmatrix} 3 \times 0 + 14 \times 6 \\ 9 \times 0 + 5 \times 6 \end{bmatrix} \pmod{26} = \begin{bmatrix} 84 \\ 30 \end{bmatrix}$$

$$\pmod{26} = \begin{bmatrix} 6 \\ 4 \end{bmatrix}$$

plaintext : TOPSECRET MESSAGE

2. Consider the Hill cipher key matrix

$$K = \begin{bmatrix} 24 & 19 \\ 5 & 14 \end{bmatrix}, \text{ cipher text : R X A O D Y} \text{ find}$$

the plain text for above cipher text.

Solⁿ.

$$\begin{bmatrix} 24 & 19 \\ 5 & 14 \end{bmatrix} = (24 \times 14) - (5 \times 19) = 241 \div 26$$

$$q \text{ rem } r = 26 \times 9 = 234 - 241 = 7 \pmod{26}$$

$$7^{-1} \pmod{26} \Rightarrow 26 = 7(3) + 5, \quad 5 = 26 - 7(3)$$

$$7 = 5(1) + 2, \quad 2 = 7 - 5(1)$$

$$5 = 2(2) + 1$$

$$1 = 5 - 2(2) = 5 - 4$$

$$1 = 5 - 2(7 - 5(1))$$

$$1 = 3(5) - 2(7)$$

$$1 = 3(26 - 7(3)) - 2(7)$$

$$1 = (-11)7 + 3(26) \Rightarrow -11 + 26 = 15$$

$$15 \begin{bmatrix} 14 & -19 \\ -5 & 24 \end{bmatrix} = \begin{bmatrix} 210 & -285 \\ -75 & 360 \end{bmatrix} \pmod{26} = \begin{bmatrix} 2 & 1 \\ 3 & 21 \end{bmatrix}$$

$$\text{ii) } \begin{bmatrix} 2 & 1 \\ 3 & 21 \end{bmatrix} \begin{bmatrix} 17 \\ 23 \end{bmatrix} = \begin{bmatrix} 2 \times 17 + 1 \times 23 \\ 3 \times 17 + 21 \times 23 \end{bmatrix} = \begin{bmatrix} 547 \\ 557 \end{bmatrix} \pmod{26}$$

$$\begin{bmatrix} 5 \\ 11 \end{bmatrix} = \begin{bmatrix} F \\ L \end{bmatrix}$$

$$\text{iii) } \begin{bmatrix} 2 & 1 \\ 3 & 21 \end{bmatrix} \begin{bmatrix} 0 \\ 14 \end{bmatrix} = \begin{bmatrix} 2 \times 0 + 1 \times 14 \\ 3 \times 0 + 21 \times 14 \end{bmatrix} = \begin{bmatrix} 14 \\ 308 \end{bmatrix} \pmod{26}$$

$$\begin{bmatrix} 14 \\ 22 \end{bmatrix} = \begin{bmatrix} O \\ W \end{bmatrix}$$

$$10) \begin{bmatrix} 2 & 1 \\ 3 & 22 \end{bmatrix} \begin{bmatrix} 3 \\ 24 \end{bmatrix} = \begin{bmatrix} 2 \times 3 + 1 \times 24 \\ 3 \times 3 + 22 \times 24 \end{bmatrix} = \begin{bmatrix} 30 \\ 537 \end{bmatrix} \text{ mod } 26$$

$$\begin{bmatrix} 4 \\ 14 \end{bmatrix} = \begin{bmatrix} e \\ n \end{bmatrix} \text{ plain text FLOWER}$$

3. Relationship b/w Substitution, Confusion, Transposition & Diffusion.

1. Substitution $\rightarrow$ Confusion.

Substitution replaces one character with another character.

Ex:- HELLO $\rightarrow$ KHOOR.

The relationship is:

Substitution creates confusion, by hiding the relationship b/w the original plaintext characters & the ciphertext characters.

So :

Substitution $\rightarrow$ Confusion.

2. Transposition $\rightarrow$ Diffusion.

Transposition changes the position/order of characters without changing the characters themselves.

Ex:-

HELLO $\rightarrow$ LHOEL.

The relationship is:

Transposition creates diffusion by spreading the influence of plaintext characters across different position in the ciphertext.

Transposition $\rightarrow$ Diffusion.

M T W T F S S
☐ ☐ ☐ ☐ ☐ ☐ ☐

COMPASS

Date :

Classical Ciphers - Confusion & Diffusion.

Cipher	Does it change the character/ Symbol value	Does it change the character position	Confusion/ Diffusion
Shift Cipher	Yes	No	Confusion
Multiplicative cipher	Yes	No	Confusion
Affine cipher	Yes	No	Confusion
Transposition cipher	No	Yes	Diffusion

Justification.

Shifts each character to another character value, but keeps its its position unchanged.

Multiplies the numerical value of each character, changing the symbol but not its position.

Applies multiplication & addition to each character value changing the symbol while keeping its position.

Keeps the character values unchanged but rearranges their position, spreading the plaintext pattern.

47 Transposition:

~~Justification.~~

Encryption:- The arrow is pointing north direction " using.

- Simple with Key = 4

- Keyword with ~~H~~ = ATLAS.

- Double Key 1 = Boreas

Key 2 = Log.

1. THE ARROW IS POINTING NORTH DIRECTION.

1. Simple Transposition - Key = 4.

T H E A

R R O W

I S P O

Column = 1

I N T I

T R I I W R I T

N G N O

R T H O

column 2:

I R E C

H I S W G T R I

T I O N

Column 3:

E O P T N H E O.

Column 4:

A W O I O D C N.

Ciphertext: ~~TRIN~~

T R I I N R I T H R S N G T R I E O P T N H E O A
 - W O I O D C N.

M T W T F S S
☐ ☐ ☐ ☐ ☐ ☐ ☐

COMPASS

Date :

b) Keyword: ATLAS

msg :- The arrow is pointing north direction.
 - Remove Spain.

- Keyboard: - A T L A S

1 5 3 2 4

t h e a r

n o w i s

p o i n t

i n g n o

1 → 2 → 3 → 4 → 5

A → A → L → S → T

1 : A

T R P I R R O

1 : A

A I N N D T

3 : L

E W I G H C

4 : 5

R S T O I

5 : T

H O O N T E N

combine:-

TRPIRR OAI NND T EWIGH CRS TOIHOON
 TEN.

Cipher ↑

c) double : Key 1 : BREAK.

Key 2 : LOG.

Key 1 : B R E A K.

2 5 3 1 4

T H e A R

R O W I S

P O I N T

I N G N O

M T W T F S S
☐ ☐ ☐ ☐ ☐ ☐ ☐

COMPASS

Date: _____

R T H D I

1 → A : A I N N D T

I R E C T I

2 → B : T I R P I R R O

O N X Y X

3 → E : E W I G H C

4 → K : R S T O I

5 → R : H O O N T E N

1st Transposition :-

A I N N D T I R P I R R O E W I G H C R S T O I
 H O O N T E N

Key 2 :-

L O G

2 3 1

A I N

N D T

T R P

I R R

O E W

I G H

C R S

T O I

H O O

N T E

N

1 : G N T R R W H S I O E

2 : L A N T I O E C T H N N

3 : O I D R R E G R O O T

N T P R W H S I O E A N T I O T

C T H N N I D R R E G R O O T

M T W T F S S
☐ ☐ ☐ ☐ ☐ ☐ ☐

COMPASS

Date : _____

5. Simple Transposition Decryption.
 Key = 10.

Ciphertext : TIDIE MANME REOGO SANOW
 RSLTP SSSNU AHTUD BIENP GODAE
 PEIXD ELNSX.

Step 1 : Find row & column
 $55 \div 10 = 5$.

Step 2 :
 T R I T N
 M R N M E
 A L O G O
 S A N O W
 R S L T P
 E E E D O
 S S S N U
 A H T U D
 B I E N P
 G O D A E
 P E I X D
 E L N S X

Step 2 : Fill the ciphertext column-wise.

T R O O P S H E A D
 I N G W E S T N E L
 O M O R E S U P E N
 I E S L I E D G I S
 E R A T I O N B O X X
 M E N O O A I D X X.

→ Read row-wise.

TROOPS HEADING
 WEST NEED MORE
 SUPPLIES SEND
 GENERAL DUBOIS
 MEN TO AID X

plaintext: TROOPS HEADING WEST NEED
 MORE SUPPLIES SEND GENERAL.
 DUBOIS MEN TO AID

67. Permutation: means rearranging a set of
 items into a particular order.

23 position 1 2 3 4 5
 1.

4	3	1	5	2
---	---	---	---	---

⇒ 3 5 2 1 4

2. 7 4 1 3 5 2 6.

position 1 2 3 4 5 6
 permutation 7 4 1 3 5 2.

⇒ 3 6 4 2 5 7 1

3. 4 1 3 2.

position 1 2 3 4
 permutation 4 1 3 2.

⇒ 2 4 3 1

4. 3 5 1 4 2.

position 1 2 3 4 5
 permutation 3 5 1 4 2.

⇒ 3 5 1 4 2